

How OUTPUT TECHNOLOGIES Are Changing For The Better



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3D printers are penetrating into the market, enabling rapid prototyping of realistic 3D models. Ever wondered how these printers are designed? In an era of unified communications, where everything is smart and connected, the prevalent output technologies are no less than a communion of traditional design and modern components. The coming together of these two results in disruptive innovations and solutions.

Take, for instance, digital speakers. Their sound output is much better than of traditionally used amplifying systems. How are devices becoming more powerful in functionality while reducing in volume? The answer is simple: The design to build them is transforming.

Here's a look at how video, audio, print and tactile designs for output technologies are being upgraded in the smart world.

Video/display

Next-generation display technologies include organic light-emitting diode (OLED), surface-conduction electron-emission display (SED), field-emission display (FED) and quantum-dot display (QD LED). Companies like Sony, Pana-

sonic, Samsung and LG are using OLED in their TV sets, smartphones and other devices. Canon and Toshiba are implementing SED, while Motorola is working on FED.

QD LED has found wide acceptance as it is reliable and energy-efficient. It can be fabricated using either of two techniques: phase separation or contact printing.

BenQ is using QD LED displays in its latest range of products. "The world is moving towards product specialisation. BenQ's 69cm (27-inch) QHD (2560x1440) IPS LED display delivers 100 per cent sRGB colour accuracy and wide viewing angle performance," shares Rajeev Singh, BenQ India managing director.

BenQ also has a range of high-tech dustproof projectors for schools. These projectors are equipped with Dust Guard technology, enabling them to last longer and reduce downtime while still delivering good image quality and performance. Dustproof line products come equipped with sealed engines, anti-dust accumulation sensor, Hall sensors, and dust filters to prolong service and ultimately save schools the trouble of cleaning, repairing and replacing dust-damaged AV equipment."

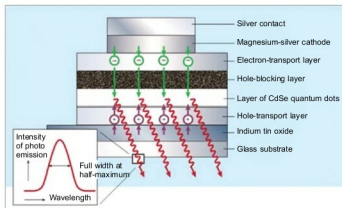


Fig. 1: QD-LED display design (Image source: www.htforum.com)

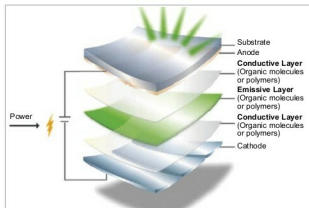


Fig. 2: OLED display design (Image source: www.elprocus.com)